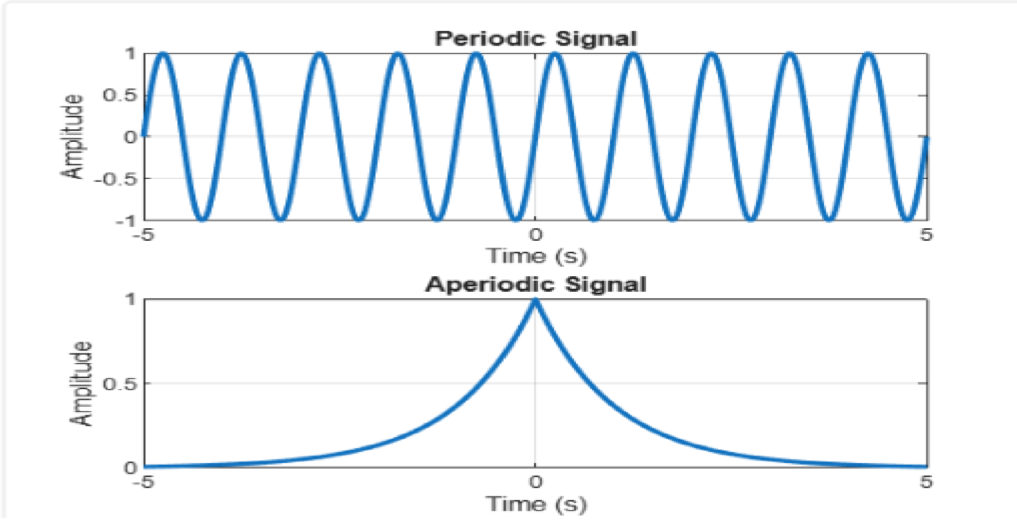
2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Experiment 2.1

```
% Periodic and Aperiodic Signals

clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Periodic Signal (Sine Wave)
x1 = sin(2*pi*t);
% Aperiodic Signal (Exponential Decay)
x2 = exp(-abs(t));
% Plotting
figure;
subplot(2,1,1);
plot(t,x1,'LineWidth',2);
grid on;
title('Periodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,x2,'LineWidth',2);
grid on;
title('Aperiodic Signal');
xlabel('Time (s)');
```

ylabel('Amplitude');



Experiment 2.2

% Verification of Even and Odd Signal Decomposition

clc;

clear;

close all;

% Time axis

t = -5:0.1:5;

% Original Signal

x = t.^3 + t;

% Time-reversed Signal

x_rev = (-t).^3 + (-t);

% Even Component

xe = (x + x_rev)/2;

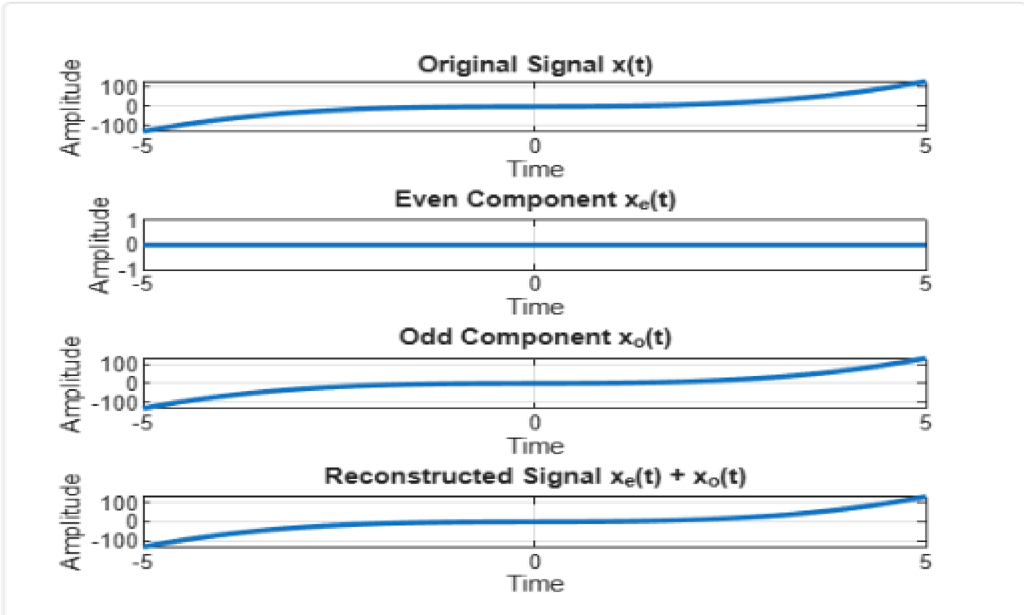
% Odd Component

xo = (x - x_rev)/2;

```
% Reconstruction of Original Signal
x_reconstructed = xe + xo;
% Plotting
figure;
subplot(4,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal x(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,2);
plot(t,xe,'LineWidth',2);
grid on;
title('Even Component x_e(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,3);
plot(t,xo,'LineWidth',2);
grid on;
title('Odd Component x_o(t)');
xlabel('Time');
ylabel('Amplitude');

subplot(4,1,4);
plot(t,x_reconstructed,'LineWidth',2);
grid on;
```

```
title('Reconstructed Signal x_e(t) + x_o(t)');
xlabel('Time');
ylabel('Amplitude');
```



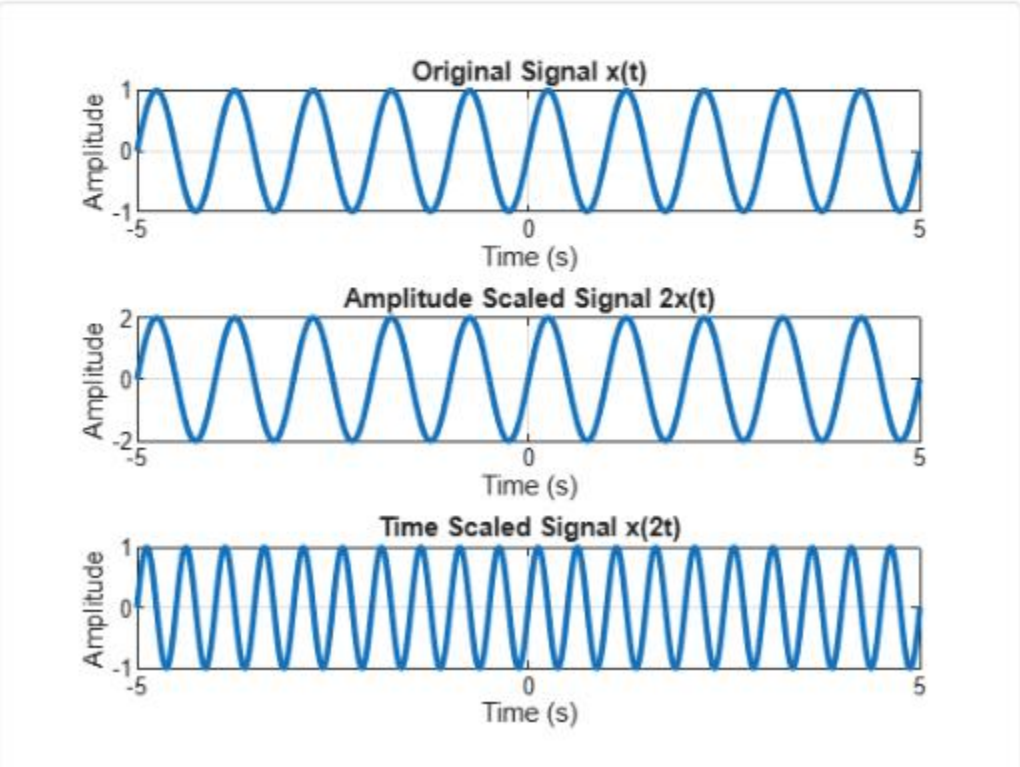
Experiment 2.3

```
% Amplitude and Time Scaling of Signals
clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Original Signal
x = sin(2*pi*t);
% Amplitude Scaling
x_amp = 2*x;
% Time Scaling (Compression)
x_time = sin(2*pi*2*t);
% Plotting
```

```
figure;
subplot(3,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,2);
plot(t,x_amp,'LineWidth',2);
grid on;
title('Amplitude Scaled Signal 2x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t,x_time,'LineWidth',2);
grid on;
title('Time Scaled Signal x(2t)');

xlabel('Time (s)');
```

ylabel('Amplitude');



Experiment 2.4

% Time Shifting and Time Reversal of Signals

clc;

clear;

close all;

% Time axis

t = -5:0.01:5;

% Original Signal

x = sin(2*pi*t);

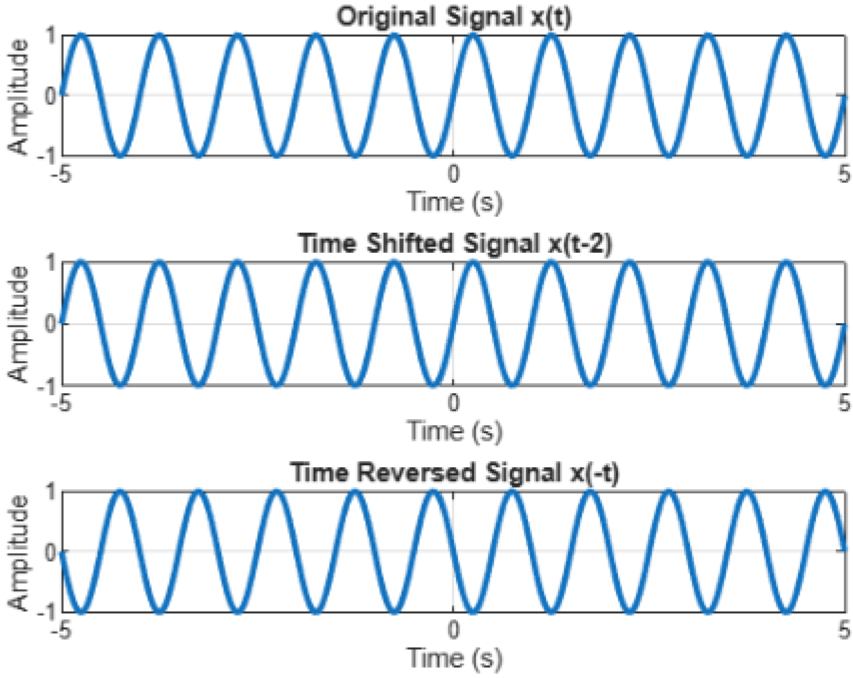
% Time Shift (Delay by 2 seconds)

x_shift = sin(2*pi*(t-2));

% Time Reversal

x_reverse = sin(2*pi*(-t));

```
% Plotting
figure;
subplot(3,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,2);
plot(t,x_shift,'LineWidth',2);
grid on;
title('Time Shifted Signal x(t-2)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t,x_reverse,'LineWidth',2);
grid on;
title('Time Reversed Signal x(-t)');
xlabel('Time (s)');
ylabel('Amplitude');
```



Original Signal x(t)
Amplitude
1
0
-1
-5
0
5
Time (s)
Time Shifted Signal x(t-2)
Amplitude
1
0
-1
-5
0
5
Time (s)
Time Reversed Signal x(-t)
Amplitude
1
0
-1
-5
0
5
Time (s)